Activity 1: Understand the correlation among Hardware, Networking, and Operating Systems.

Basic (1-10)

Q: What is hardware?

A: The physical, tangible components of a computer system (e.g., motherboard, CPU, RAM).

Q: What is an Operating System (OS)?

A: Software that manages computer hardware and software resources and provides common services for computer programs (e.g., Windows, Linux).

Q: What is networking?

A: The practice of connecting computers and other devices to share resources and information.

Q: Which component executes the instructions of a computer program?

A: The Central Processing Unit (CPU).

Q: Name a common operating system.

A: Microsoft Windows.

Q: What does RAM stand for?

A: Random Access Memory.

Q: What is the primary purpose of a network?

A: To enable communication and resource sharing between devices.

Q: Which software acts as an intermediary between the user and the hardware?

A: The Operating System.

Q: True or False: A computer can function without an operating system.

A: False. While possible for very specific embedded systems, for all practical purposes, a general-purpose computer requires an OS to be usable.

Q: What hardware component connects all other components in a computer?

A: The Motherboard.

Intermediate (11-15)

11. Q: How does the OS interact with hardware?

A: Through device drivers, which are software programs that allow the OS to communicate with specific hardware components.

12 Q: Why is a network card essential for networking?

A: It provides the physical interface and hardware address (MAC address) needed to connect a computer to a network.

13 Q: Explain how the OS, hardware, and a web browser work together when you open a website.

A: You click the browser icon (Application). The OS (Windows/Linux) loads the browser into RAM (Hardware) and manages its access to the CPU. The browser requests data via the Network Card (Hardware), which is managed by the OS's networking stack.

14 Q: What role does the OS play in managing RAM?

A: It allocates memory to running applications, manages virtual memory, and protects memory spaces to prevent one application from crashing another.

15 Q: Can a networking problem be caused by faulty hardware? Give an example.

A: Yes. A faulty network cable, a damaged network interface card (NIC), or a misconfigured router can all cause networking issues.

Advanced (16-20)

16. Q: Describe the boot process and the roles of hardware (BIOS/UEFI) and the OS.

A: The CPU loads the BIOS/UEFI (firmware on a hardware chip). BIOS/UEFI performs a Power-On Self-Test (POST), initializes hardware, and locates the bootloader on the storage device. The bootloader then loads the OS kernel into RAM, and the OS takes over control.

17 Q: How does an OS's network stack abstract the underlying network hardware from applications?

A: It provides a standardized API (like sockets) for applications to use. The application doesn't need to know if the connection is over Ethernet or Wi-Fi; the OS's stack handles the hardware-specific details.

18 Q: What is the relationship between a device driver, the OS kernel, and hardware?

A: The device driver is a kernel module that translates generic OS commands (e.g., "write this data to disk") into specific commands that the particular hardware (e.g., a specific SSD model) understands.

19 Q: Explain how a virtual machine relies on the correlation between hardware, OS, and a hypervisor.

A: The hypervisor (a specialized OS or software) runs directly on the physical hardware. It virtualizes the hardware (vCPU, vRAM, vNIC) and presents it to the guest OS. The guest OS then interacts with this virtual hardware as if it were real, unaware it's being shared.

20 Q: In a distributed system, how do hardware failure, network partitioning, and the OS's fault tolerance mechanisms correlate?

A: The system's reliability depends on this correlation. If hardware fails, the OS on other nodes must detect it (via network timeouts). Network partitioning can create split-brain scenarios. Advanced OS and application logic are needed to maintain data consistency and availability despite these hardware and network issues.

Activity 2: Assembling various Hardware like Motherboard, RAM, ROM, SMPS, I/O Ports, etc.

Basic (1-10)

1 Q: What is the main circuit board of a computer called?

A: The Motherboard.

2 Q: What does RAM do?

A: It provides fast, temporary storage for data and programs that are currently in use.

3 Q: What is the purpose of the SMPS?

A: The Switched-Mode Power Supply (SMPS) converts AC power from the wall outlet to the lower DC voltages required by the computer's internal components.

4 Q: Where is the BIOS/UEFI firmware stored?

A: On a ROM (Read-Only Memory) or, more commonly, a flash memory chip on the motherboard.

5 Q: Name two common types of RAM modules.

A: DIMM (for desktops) and SODIMM (for laptops).

6 Q: What should you do to protect components from Electrostatic Discharge (ESD) before handling them?

A: Wear an anti-static wrist strap and work on an anti-static mat.

7 Q: Which component is considered the "brain" of the computer?

A: The Central Processing Unit (CPU).

8 Q: What is the primary function of a heat sink and fan on the CPU?

A: To dissipate heat and prevent the CPU from overheating.

9 Q: Which port is typically used for connecting a monitor?

A: HDMI, DisplayPort, or VGA.

10 Q: What is the purpose of the I/O shield?

A: It is a metal plate that fits into the back of the computer case, providing holes for the motherboard's I/O ports and shielding against EMI.

Intermediate (11-15)

11. Q: Why is it crucial to match the CPU socket type on the motherboard with the CPU itself?

A: The physical and electrical connections are specific to a generation or brand of CPUs. A mismatch will make installation impossible and could damage the components.

12 Q: What is the significance of the notch in a RAM module?

A: It is a keying mechanism to ensure the module is inserted in the correct orientation and to prevent incompatible RAM types (e.g., DDR4 vs. DDR5) from being installed.

13 Q: When connecting the front panel connectors (power button, LEDs), what is your primary reference?

A: The motherboard's manual, as the pin layouts are not standardized and vary between manufacturers.

14 Q: What is the difference between SATA and NVMe storage?

A: SATA is an older interface for connecting storage like HDDs and 2.5" SSDs. NVMe is a newer, faster protocol that connects SSDs directly to the CPU via the PCIe bus, bypassing the SATA bottleneck.

15 Q: How do you determine the correct orientation for installing the CPU?

A: Look for a small triangle or notch on one corner of the CPU and a corresponding marker on the CPU socket.

Advanced (16-20)

16. Q: Describe the process of updating the BIOS/UEFI and why it might be necessary for a new build.

A: It is done by downloading a firmware file from the manufacturer onto a USB drive and using the motherboard's built-in flash utility. It's often necessary to support newer CPU generations or to fix critical bugs.

17 Q: What is the role of the chipset on the motherboard, and how does it relate to the CPU?

A: The chipset manages data flow between the CPU, RAM, storage, and peripheral buses (like SATA and USB). Modern CPUs integrate many of these functions, making the chipset a high-speed I/O controller.

18 Q: Explain the concept of dual-channel memory configuration. How do you physically install RAM to enable it?

A: It doubles the communication rate between the memory controller and RAM. You install matching RAM pairs in specific, same-colored slots on the motherboard as per the manual (e.g., slots A2 and B2).

19 Q: When troubleshooting a "no POST" situation after assembly, what are the first components you would check?

A: 1. All power connections (24-pin ATX, 8-pin CPU). 2. CPU installation. 3. RAM (reseat it, try one stick at a time). 4. Front panel power switch connection.

20 Q: What is the difference between a modular and non-modular power supply, and what is the advantage of a modular one during assembly?

A: A non-modular PSU has all cables permanently attached. A modular PSU allows you to connect only the cables you need, resulting in significantly better cable management and improved airflow inside the case.

Activity 3: Installation of Operating Systems in Computers.

Basic (1-10)

1. What is an OS? (Software that manages hardware/software resources).

2. Name two common desktop OSes. (Windows, macOS, Linux).

3. What is an ISO file? (A disk image file containing all the data from an optical disc).

4. How do you typically start an OS installation? (Boot from a USB drive or DVD).

5. What is partitioning? (Dividing a hard drive into separate, logical sections).

6. What is the "C:" drive typically? (The primary partition where Windows is installed).

7. What does "formatting a drive" mean? (Preparing it for use by an OS, which erases all data).

8. What is a filesystem? (A method for storing and organizing files - e.g., NTFS, FAT32, ext4).

9. What is a device driver? (Software that allows the OS to communicate with hardware).

10. What is the first screen you see after a successful Windows boot? (The Lock Screen/Login Screen).

Intermediate (11-15)

11. What is the difference between MBR and GPT partition schemes?

A: MBR is older, supports up to 2TB disks, and has 4 primary partition limits. GPT is newer, supports larger disks, and has virtually no partition limit. It's required for UEFI boot.

12. Why might you need to load third-party drivers during OS installation?

A: If the installation media doesn't have drivers for critical hardware like storage controllers (e.g., RAID) or network cards.

13. What is the purpose of a bootloader (e.g., GRUB, Windows Boot Manager)?

A: To load the OS kernel into memory and hand over control to it. It also allows for dual-booting multiple OSes.

14. What happens during the "Installing Updates" phase of a Windows setup?

A: The installer downloads and integrates the latest security patches and critical fixes into the OS image before the first boot.

15 Q. What is the difference between an "Upgrade" and a "Custom" (clean) installation?

A: Upgrade preserves files, settings, and applications. Custom install erases everything on the partition, providing a fresh start.

Advanced (16-20)

16. Describe the process of creating a bootable USB drive from an ISO file.

A: Use a tool like Rufus or the Windows Media Creation Tool. The tool writes the ISO's contents to the USB drive and makes it bootable by setting up a master boot record and copying necessary system files.

17. What are "answer files" (e.g., autounattend.xml for Windows) used for?

A: They automate the OS installation process by providing pre-configured answers to setup prompts, enabling unattended deployments.

18. Explain the concept of "PXE boot" and how it's used for OS deployment over a network.

A: Preboot Execution Environment. A client computer boots and requests an OS image from a network server, allowing for centralized, automated deployment without physical media.

19. What is the Windows Recovery Environment (WinRE) and when is it used?

A: A separate, minimal OS used to troubleshoot and repair a Windows installation that fails to start, allowing for system restore, startup repair, and command-line access.

20. How does a Linux distribution's package manager (e.g., apt, yum) relate to the OS installation and post-installation setup?

A: The installer itself uses the package manager to install core packages. After installation, the user uses the same tool to install, update, and remove all other software, ensuring dependency management and system consistency.

Activity 4: Installation of Virtual Box and it's working.

Basic (1-10)

1. What is a virtual machine (VM)?

A: A software emulation of a physical computer that runs its own OS.

2. What is a hypervisor (or Virtual Machine Monitor)?

A: Software that creates and runs virtual machines (Type 2: e.g., VirtualBox; Type 1: e.g., VMware ESXi).

3. What is an Oracle VM VirtualBox?

A: A free and open-source hosted (Type 2) hypervisor for x86 virtualization.

4. What is a "Guest OS"?

A: The operating system that runs inside a virtual machine.

5. What is the "Host OS"?

A: The operating system running directly on the physical hardware where the virtualization software is installed.

6. Name a file format for a virtual hard disk in VirtualBox.

A: VDI (VirtualBox Disk Image), VMDK, VHD.

7. What are "Guest Additions"?

A: Special software packages installed inside the guest OS to improve performance and enable features like shared folders and seamless mouse integration.

8. What does "snapshot" do in VirtualBox?

A: It saves the current state of a VM (disk, memory, settings) allowing you to revert to it later.

9. What is the purpose of allocating VRAM to a VM?

A: To provide video memory for the emulated graphics card, improving display performance.

10. What is NAT network mode in VirtualBox?

A: The default mode where the VM shares the host's IP address, can access the external network, but is not directly accessible from the network.

Intermediate (11-15)

11. What is the difference between "Bridged" and "NAT" networking?

A: In Bridged mode, the VM gets its own IP on the physical network, like a separate physical machine. In NAT, it's hidden behind the host's IP.

12. What is a "Shared Folder" and how is it set up?

A: A folder on the host OS that can be accessed from within the guest OS. It is set up via the VM's settings and requires Guest Additions.

13. Why would you need to enable Intel VT-x or AMD-V in the BIOS?

A: These are hardware virtualization extensions that significantly improve VM performance and are required for 64-bit guest OSes.

14. What is the purpose of the "Host-Only" network adapter?

A: It creates a private network between the host OS and its VMs, but the VMs cannot access the external internet.

15. Explain the concept of "Cloning" a VM.

A: It creates a copy of an existing VM. A "Full Clone" is independent, while a "Linked Clone" depends on the original VM's disk.

Advanced (16-20)

16. Describe how you would use the VBoxManage command-line tool to create and manage a VM.

A: VBoxManage createvm --name "MyVM" --register then VBoxManage modifyvm "MyVM" --memory 1024 --cpus 2 etc. It allows for automation and scripting of VM management.

17. What is the difference between a "Dynamically Allocated" and "Fixed Size" virtual hard disk?

A: Dynamic starts small and grows as the guest OS uses more space. Fixed allocates the entire disk size immediately, which is slower to create but can offer better performance.

18. How does "Seamless Mode" work and what enables it?

A: It allows you to run guest application windows directly on the host desktop, blurring the line between host and guest. It is enabled by the Guest Additions.

19. Explain a use case for an "Internal Network" configuration.

A: To create an isolated software-defined network between multiple VMs on the same host, ideal for building virtual labs for testing networking, security, or clustered software without any connection to the host or external network.

20. What are the security implications of using "Bridged" networking for a VM?

A: The VM is exposed on the local network and is as vulnerable as a physical machine. It must have its own firewall and security updates, as it can be directly targeted by attackers.

Activity 5: Experiment with drivers and application software's installation and uninstallation.

Basic (1-10)

1. What is a device driver?

A: Software that allows the OS to communicate with a specific hardware device.

2. Where can you view installed devices and their drivers in Windows?

A: Device Manager.

3. What is the most common file type for a Windows application installer?

A: .exe (executable) or .msi (Windows Installer Package).

4. What is the primary way to uninstall a program in Windows?

A: Through "Apps & features" in Settings or "Programs and Features" in the Control Panel.

5. What does "Plug and Play" (PnP) mean?

A: The OS automatically detects a new hardware device and attempts to install a driver for it.

6. What is a driver signature?

A: A digital certificate that verifies the driver publisher and ensures the driver has not been tampered with.

7. Where does a 64-bit Windows typically store 64-bit application files?

A: In the C:\Program Files directory.

8. What is the purpose of an application's "setup.exe" or "install.exe" file?

A: To launch the installation wizard for that application.

9. What is a common symptom of a missing or corrupt driver?

A: The device does not work at all, or it appears with a yellow exclamation mark in Device Manager.

10. What is the purpose of an "Express" vs. "Custom" installation?

A: Express installs with default settings, often including bundled software. Custom allows you to choose installation location and components.

Intermediate (11-15)

11. Why might you need to "Roll Back" a driver?

A: If a newly updated driver is causing system instability or device failure, rolling back reverts to the previous, working version.

12. What is the difference between installing from an .msi file versus a standalone .exe?

A: .msi files are Windows Installer packages that offer standardized installation, repair, and uninstallation, and can be deployed across a network. .exe files are often self-extracting archives with custom installers.

13. How can you manually update a driver in Windows?

A: Right-click the device in Device Manager -> "Update driver" -> "Browse my computer for drivers".

14. What are "residual files" after uninstallation?

A: Leftover files, folders, or registry entries that were not removed by the program's uninstaller.

15. What is a "driver store" in Windows?

A: A protected location (C:\Windows\System32\DriverStore) where Windows keeps a cache of all installed driver packages.

Advanced (16-20)

16. What is the purpose of the "pnputil.exe" command-line tool?

A: It is used to manage the driver store, allowing you to add, remove, and list driver packages from the command line.

17. Explain a scenario where you would need to "force install" an unsigned driver and the risks involved.

A: When using very new or niche hardware that doesn't have a signed driver. The risk is that the driver could be malicious or unstable, potentially causing system crashes.

18. How does the Windows Registry play a role in software installation?

A: It stores configuration settings, file associations, and uninstallation information for installed applications.

19. What are "silent" or "unattended" installation parameters (like /S or /quiet)?

A: Command-line switches used with installers to run them without user interaction, which is useful for scripting mass deployments.

20. Describe the relationship between a driver (kernel mode) and an application (user mode).

A: Applications run in a protected, non-privileged "user mode." When an app needs to access hardware, it makes a system call to the OS, which then passes the request to the driver running in privileged "kernel mode," which has direct hardware access.

Activity 6: Make use of Crimping Tool to design LAN Cable...

Basic (1-10)

1. What is a LAN cable commonly known as?

A: An Ethernet cable or Twisted Pair cable.

2. What is the most common standard for Ethernet cables today?

A: CAT5e or CAT6.

3. What is the tool called used to attach connectors to a cable?

A: A Crimping Tool.

4. What is the name of the connector at the end of an Ethernet cable?

A: An RJ-45 connector.

5. What are the two main wiring standards for Ethernet cables?

A: T568A and T568B.

6. Which standard is more commonly used in new installations?

A: T568B.

7. What is a "straight-through" cable used for?

A: Connecting different types of devices (e.g., PC to Switch, PC to Router).

8. What is a "crossover" cable used for?

A: Connecting similar devices (e.g., Switch to switch, PC to PC).

9. What tool is used to test a newly crimped cable?

A: A LAN Cable Tester.

10. What is the purpose of the plastic jacket on a cable?

A: To protect the internal wires from physical damage.

Intermediate (11-15)

11. How many pairs of wires are in a standard CAT5e/6 cable?

A: 4 pairs (8 wires total).

12. What is the pinout order for T568B?

A: 1. White-Orange, 2. Orange, 3. White-Green, 4. Blue, 5. White-Blue, 6. Green, 7. White-Brown, 8. Brown.

13. Why are the wires twisted in pairs inside the cable?

A: To reduce crosstalk (interference between pairs) and electromagnetic interference (EMI) from external sources.

14. In a crossover cable, which two pairs are swapped?

A: The transmit (TX) and receive (RX) pairs. Specifically, pins 1&2 are swapped with 3&6.

15. Why are modern switches and NICs "auto-MDIX"?

A: They can automatically detect the cable type (straight-through or crossover) and adjust their transmit/receive pairs accordingly, making crossover cables largely obsolete for Ethernet.

Advanced (16-20)

16. Describe the step-by-step process of crimping an RJ-45 connector onto a CAT6 cable.

A: 1. Strip the outer jacket. 2. Untwist and arrange wires in T568B order. 3. Trim wires evenly. 4. Insert wires fully into RJ-45 connector (jacket should enter the connector). 5. Insert connector into crimping tool and squeeze firmly.

17. What is Power over Ethernet (PoE) and how does it affect cable standards?

A: A technology that allows electrical power to be transmitted along with data on Ethernet cables. It requires all 4 pairs in higher-power versions, making CAT5e the minimum and CAT6/6A recommended.

18. What is the difference between a solid core and a stranded core Ethernet cable, and where is each typically used?

A: Solid core is rigid and used for permanent installations (inside walls). Stranded core is flexible and used for patch cables (connecting devices to wall jacks).

19. Explain the purpose of the "sheath" or "boot" on an RJ-45 connector.

A: To protect the plastic clip from snapping off and to provide strain relief for the cable.

20. What would you check if a crimped cable fails the cable tester on specific pins?

A: 1. the wire order is incorrect. 2. The wires are not making contact with the pins (not pushed in far enough before crimping). 3. The crimp is faulty and needs to be redone.

Activity 7: Apply the concept of network to configure the IPV4 for private and public

Basic (1-10)

1. What is an IP address?

A: A unique numerical label assigned to each device connected to a network that uses the Internet Protocol for communication.

2. What is the difference between a private and a public IP address?

A: A public IP is globally routable on the internet. A private IP is used within a local network (LAN) and is not directly accessible from the internet.

3. What is the purpose of a Subnet Mask?

A: To distinguish the network portion and the host portion of an IP address.

4. What is a default gateway?

A: The IP address of the router on your local network that connects you to other networks (e.g., the internet).

5. What does DHCP stand for?

A: Dynamic Host Configuration Protocol.

6. What is the function of DHCP?

A: To automatically assign IP addresses, subnet masks, default gateways, and other network configuration parameters to devices on a network.

7. Name one RFC 1918 private IP address range.

A: 10.0.0.0 - 10.255.255.255, 172.16.0.0 - 172.31.255.255, or 192.168.0.0 - 192.168.255.255.

8. What command is used to check your IP configuration in Windows?

A: ipconfig.

9. What is the loopback IP address?

A: 127.0.0.1 (refers to the local computer itself).

10. What is APIPA?

A: Automatic Private IP Addressing (169.254.x.x), an address self-assigned when a device cannot get an IP from a DHCP server.

Intermediate (11-15)

11. How do you manually set a static IP address in Windows?

A: Network and Sharing Center -> Change adapter settings -> Right-click adapter -> Properties -> Internet Protocol Version 4 (TCP/IPv4) -> Properties -> Use the following IP address.

12. Why would you assign a static IP address to a network printer?

A: So its address never changes, making it reliably accessible to all computers on the network.

13. What is the relationship between a public IP on a router and the private IPs on devices behind it?

A: The router uses Network Address Translation (NAT) to allow multiple devices with private IPs to share the single public IP for internet communication.

14. What is DNS and why is it needed even if you have an IP address?

A: Domain Name System. It translates human-friendly domain names (e.g., google.com) into machine-readable IP addresses. We use names because they are easier to remember than numbers.

15. What is the difference between ipconfig /release and ipconfig /renew?

A: /release tells the DHCP server "I no longer need this IP address." /renew requests a new lease from the DHCP server.

Advanced (16-20)

16. Given an IP address of 192.168.1.50 and a subnet mask of 255.255.255.0, what is the network address and the broadcast address?

A: Network: 192.168.1.0, Broadcast: 192.168.1.255.

17. Explain how NAT (Network Address Translation) works.

A: The router maintains a translation table. When a device with a private IP sends a packet to the internet, the router replaces the source private IP with its own public IP and a unique port number. When a response comes back, the router uses the port number to forward the packet to the correct private device.

18. What is CIDR notation? Represent 255.255.255.0 in CIDR.

A: Classless Inter-Domain Routing. It represents an IP address and its subnet mask with a prefix. 255.255.255.0 is /24.

19. What is the purpose of a reservation in a DHCP server?

A: To always assign the same specific IP address to a particular device (identified by its MAC address), combining the convenience of DHCP with the permanence of a static IP.

20. Describe a scenario where you might use a custom subnet mask (like 255.255.255.128) on a small LAN.

A: To divide a single network (e.g., 192.168.1.0/24) into two smaller subnets for different departments or for segregating guest WiFi from the main network, improving security and reducing broadcast traffic.

Activity 8: Demonstrate the concept of RDP for remotely controlling another computer…

Basic (1-10)

1. What does RDP stand for?

A: Remote Desktop Protocol.

2. What is the primary use of RDP?

A: To remotely control and interact with the graphical desktop of another computer over a network.

3. Which Windows editions include the RDP server functionality (can be controlled)?

A: Windows Pro, Enterprise, and Education. (Windows Home can only be a client).

4. What is the name of the built-in RDP client in Windows?

A: Remote Desktop Connection (mstsc.exe).

5. What information do you need to start an RDP session?

A: The IP address or hostname of the target computer and a user account with permission to log on.

6. Name an alternative to RDP for remote access.

A: TeamViewer, AnyDesk, Chrome Remote Desktop, VNC.

7. What is a key security consideration when using RDP over the internet?

A: It can be a target for brute-force attacks. It should be secured with strong passwords, ideally behind a VPN, or have its port changed.

8. What does "clipboard sharing" in RDP allow you to do?

A: Copy and paste text and files between the local and remote computer.

9. How can you transfer files using a basic RDP session?

A: By enabling "Drive Redirection" in the Local Resources tab, which makes your local drives appear in the remote computer's File Explorer.

10. What happens to the local session when you RDP into a Windows machine?

A: By default, the local session is locked, and the remote user takes over the active console session (or a new one).

Intermediate (11-15)

11. What port does RDP use by default?

A: TCP port 3389.

12. Where do you enable RDP on a Windows host?

A: System Properties -> Remote Desktop -> Enable Remote Desktop.

13. What are Network Level Authentication (NLA) and why is it recommended?

A: It requires the user to authenticate before a full RDP session is established, which is more secure and protects against denial-of-service attacks.

14. How can you improve RDP performance over a slow network connection?

A: In the Experience tab of the RDP client, choose a lower connection speed (e.g., "Low-speed broadband") which disables visual effects like desktop background and font smoothing.

15. What is the difference between "Admin" and "Non-Admin" users in terms of RDP connections?

A: By default, members of the "Administrators" group can RDP. Standard users must be explicitly added to the "Remote Desktop Users" group.

Advanced (16-20)

16. Explain how RDP Gateway works and its security benefit.

A: It is a role service that acts as a secure broker. Clients connect to the Gateway on port 443 (HTTPS), which then forwards the traffic to the target RDP host inside the corporate network. This means only port 443 needs to be opened on the firewall, not 3389 for every internal machine.

17. What are "RemoteApp" programs?

A: A feature that allows you to make a specific application, rather than the entire desktop, available to remote users. The app runs on the remote server but appears as if it's running on the user's local machine.

18. How can you use command-line tools to automate or troubleshoot RDP connections?

A: Use mstsc /v:IPaddress to connect. Use qwinsta and rwinsta on the host to query and reset sessions. Use netstat -an | findstr 3389 to check if the RDP port is listening.

19. Describe a scenario where you would use a VNC solution over RDP.

A: To access a computer running a non-Windows OS (like Linux) where RDP is not native, or to access the actual physical console session of a machine (e.g., to see what's on the monitor), which RDP does not do by default.

20. What is the role of digital certificates in securing RDP connections?

A: RDP uses TLS/SSL to encrypt the session. The host presents a certificate to the client to prove its identity. By default, it uses a self-signed certificate, but for higher security, a certificate from a trusted internal or public CA should be used to prevent "man-in-the-middle" attacks.

Activity 9: Plan the data backup and data restore in a system...

Basic (1-10)

1. What is data backup?

A: The process of creating a copy of data to protect against accidental loss or corruption.

2. What is the "3-2-1" backup rule?

A: Have 3 copies of your data, on 2 different media, with 1 copy stored offsite.

3. Name a location for an offsite backup.

A: Cloud storage (e.g., AWS S3, Backblaze), a bank safety deposit box, or a different physical location.

4. What is the difference between a full backup and an incremental backup?

A: Full backs up all selected files. Incremental backs up only files that have changed since the last backup (of any type).

5. What is a system image backup?

A: A complete snapshot of an entire drive, including the OS, applications, settings, and all files.

6. What is Windows built-in backup tool called?

A: File History (for files) and "Backup and Restore (Windows 7)" for system images.

7. What is a restore point in System Restore?

A: A snapshot of system files and the Windows Registry at a specific point in time, used to undo system changes.

8. What does "cloud backup" refer to?

A: Storing backup copies of data on a remote server accessed via the internet.

9. Why should you periodically test your backups?

A: To ensure they are working correctly and that data can be successfully restored.

10. What is the difference between backup and archiving?

A: Backup is for operational recovery (short-term). Archiving is for long-term retention of data that is no longer actively used.

Intermediate (11-15)

11. What is a differential backup?

A: Backs up all files that have changed since the last full backup.

12. Compare the restore process of Incremental vs. Differential backups.

A: To restore from Incremental, you need the last Full backup + all subsequent Incremental backups. For Differential, you only need the last Full + the last Differential.

13. What is the Recovery Time Objective (RTO)?

A: The target time within which a business process must be restored after a disaster to avoid unacceptable consequences.

14. What is the Recovery Point Objective (RPO)?

A: The maximum acceptable amount of data loss measured in time (e.g., if you back up every 24 hours, your RPO is 24 hours).

15. What are the advantages of using external hard drives for backup?

A: Low cost, portability, and they are air-gapped (not permanently connected, protecting against ransomware).

Advanced (16-20)

16. Describe a backup strategy for a small business using a combination of Full and Incremental backups.

A: A Full back up every Sunday night. Incremental backups every weekday night. Retain backups for 4 weeks. This balances storage space and restore complexity.

17. What is the role of VSS (Volume Shadow Copy Service) in Windows backups?

A: It allows the backup tool to take a snapshot of files that are currently open and in use (like databases), ensuring a consistent backup is created without interrupting applications.

18. Explain how versioning works in cloud backup services like Dropbox or OneDrive.

A: The service automatically keeps previous versions of files for a set period. If a file is corrupted or deleted, you can revert to a specific earlier version.

19. What is "immutable" storage and why is it important for backups?

A: A storage system where data, once written, cannot be altered or deleted for a specified retention period. This protects backups from being encrypted or deleted by ransomware.

20. How would you perform a "bare-metal recovery" of a system?

A: Boot from a recovery disk or Windows installation media, choose "Repair your computer," and then select "System Image Recovery" to restore the entire system image to a new, blank hard drive.

Activity 10: Build knowledge of the Linux File System, study and execute the following Linux and Windows Shell Commands...

Basic (1-10)

1. What is a shell?

A: A command-line interpreter that provides a user interface to access the operating system's services.

2. What command is used to create a new directory in both Linux and Windows?

A: mkdir (Make Directory).

3. What does the pwd command do in Linux?

A: Print Working Directory - shows the current directory you are in.

4. What is the Windows equivalent of pwd?

A: cd (with no arguments).

5. What does the ls command do in Linux?

A: Lists the contents of a directory.

6. What is the Windows equivalent of ls?

A: dir.

7. What does the cd command do?

A: Change Directory.

8. What does cp stand for and what is its basic function?

A: Copy. It copies files or directories.

9. What is the purpose of the mv command?

A: To move or rename files and directories.

10. What does the rm command do?

A: Removes (deletes) files.

Intermediate (11-15)

11. What is the difference between rm and rmdir?

A: rm deletes files. rmdir deletes empty directories. rm -r is used to delete directories and their contents recursively.

12. What does the touch command do?

A: It creates a new, empty file or updates the access/modification timestamp of an existing file.

13. What is the purpose of the free command in Linux?

A: To display the amount of free and used memory (RAM and swap) in the system.

14. How do you get help for a command in Linux (e.g., for mkdir)?

A: man mkdir or mkdir --help.

15. What is the difference between a relative path and an absolute path?

A: A relative path is relative to the current directory (e.g., Documents/file.txt). An absolute path starts from the root (/ in Linux, C:\ in Windows) (e.g., /home/user/Documents/file.txt).

Advanced (16-20)

16. Explain the use of the -R (or -r) flag with commands like cp, rm, and chmod.

A: It stands for "recursive," meaning the command will operate on the directory and all of its contents (subdirectories and files).

17. What does the command chmod 755 filename do?

A: It changes the file permissions. The owner gets read, write, and execute (7). The group and others get read and execute (5). (7=rwx, 5=r-x).

18. How would you use the grep command to search for the word "error" in a file called log.txt?

A: grep "error" log.txt

19. What is the purpose of the | (pipe) symbol in the command line?

A: It takes the output of the command on the left and uses it as the input for the command on the right (e.g., ls -l | grep ".txt").

20. Compare the tasklist and taskkill commands in Windows with their Linux counterparts.

A: tasklist is similar to ps aux. taskkill /PID 1234 is similar to kill 1234, and taskkill /IM notepad.exe is similar to pkill notepad.

Activity 11: Demonstrate the installation and working of Antivirus software...

Basic (1-10)

1. What is antivirus software?

A: A program designed to detect, prevent, and remove malware.

2. What is malware?

A: Malicious software (e.g., viruses, worms, trojans, ransomware).

3. Name a common antivirus program.

A: Windows Defender, Norton, McAfee, Kaspersky.

4. What is a virus signature or definition?

A: A unique string of bits that identifies a specific piece of malware, which the antivirus uses for detection.

5. What is a "full system scan"?

A: A deep scan of all files and running processes on the entire computer.

6. What is a "quick scan"?

A: A scan of critical system areas and memory where malware is most likely to be active.

7. What does "quarantine" mean in antivirus software?

A: Isolating a detected malicious file by moving it to a secure folder to prevent it from causing harm, while allowing you to restore it if it's a false positive.

8. Why is it important to keep antivirus software updated?

A: To ensure it has the latest virus definitions to detect new and emerging threats.

9. What is Windows Defender?

A: The built-in, free antivirus and anti-malware solution included with Windows 10 and 11.

10. What is a heuristic analysis?

A: A method used by antivirus software to detect new, unknown viruses by analysing code behaviour for suspicious patterns.

Intermediate (11-15)

11. What is the difference between on-access and on-demand scanning?

A: On-access (real-time) scanning checks files automatically as they are accessed. On-demand scanning is manually initiated by the user.

12. What is a "false positive"?

A: When the antivirus incorrectly flags a safe, legitimate file as being malicious.

13. What is a "false negative"?

A: When the antivirus fails to detect a piece of malware.

14. What is ransomware and how can antivirus help prevent it?

A: Malware that encrypts your files and demands a ransom. Antivirus can detect and block known ransomware strains and use behavioural analysis to stop encryption attempts.

15. Besides viruses, what other threats can modern antivirus/anti-malware software protect against?

A: Spyware, adware, rootkits, phishing attacks, and potentially unwanted programs (PUPs).

Advanced (16-20)

16. How does "cloud-based" detection enhance traditional signature-based antivirus?

A: It offloads analysis to a powerful cloud server, allowing for faster detection of new threats and reducing the resource load on the local machine.

17. What is EDR (Endpoint Detection and Response) and how does it differ from traditional antivirus?

A: EDR is a more advanced solution that not only prevents threats but also continuously monitors the endpoint for suspicious activities, records them, and provides tools for investigation and response to sophisticated attacks.

18. Explain the concept of "sandboxing" in malware analysis.

A: Running a suspicious program in an isolated, virtual environment to observe its behaviour without risking the host system.

19. Why might an organization use a centralized management console for its antivirus software?

A: To deploy, configure, update, and monitor the antivirus status on all endpoints from a single location, ensuring consistent policy enforcement and rapid response to threats.

20. What are the potential performance impacts of antivirus software, and how can they be mitigated?

A: It can slow down file access and system boot times due to real-time scanning. This can be mitigated by scheduling full scans during off-hours and excluding specific, trusted high-performance folders (like database directories) from real-time scanning.

Activity 12: Explore and configure Windows user accounts, permissions, and folder sharing in a network.

Basic (1-10)

1. Where do you create and manage user accounts in Windows?

A: In "Computer Management" (lusrmgr.msc) or "Settings" -> "Accounts" -> "Family & other users".

2. What are the two main types of Windows user accounts?

A: Administrator and Standard User.

3. What is the difference between an Administrator and a Standard User?

A: An administrator has full control over the system. A standard user can only use installed software and change non-system-wide settings.

4. What is a workgroup?

A: A simple peer-to-peer network model where each computer manages its own security.

5. What is a domain?

A: A client/server network model where security is centralized on a domain controller (e.g., Active Directory).

6. How do you share a folder with other users on the same computer?

A: Right-click folder -> Properties -> Sharing tab -> Advanced Sharing -> Share this folder.

7. How do you share a folder with users on the network?

A: Right-click folder -> "Give access to" -> "Specific people...".

8. What are NTFS permissions?

A: Permissions that control access to files and folders on an NTFS-formatted drive (e.g., Full Control, Modify, Read & Execute, Read, Write).

9. What are Share permissions?

A: Permissions that control network access to a shared folder (Read, Change, and Full Control).

10. Which is more granular, NTFS or Share permissions?

A: NTFS permissions.

Intermediate (11-15)

11. What is the principle of "least privilege"?

A: Giving a user account only the permissions absolutely necessary to perform its tasks.

12. When accessing a file over the network, how are Share and NTFS permissions combined?

A: The most restrictive permission between the two is applied.

13. What is the "Effective Access" tab in Windows used for?

A: To see the net permissions a specific user or group has on a file or folder, taking all group memberships and permissions into account.

14. What is the "Everyone" group?

A: A built-in group that includes all users, even anonymous guests.

15. Why is it bad practice to share a folder with "Everyone: Full Control"?

A: It allows any user on the network, including untrusted or malicious ones, to add, change, or delete any file in the share.

Advanced (16-20)

16. What is the difference between "Allow" and "Deny" in NTFS permissions?

A: "Allow" grants a permission. "Deny" explicitly blocks it. "Deny" overrides any "Allow" permissions.

17. How do you take ownership of a file or folder in Windows?

A: Right-click -> Properties -> Security -> Advanced -> Owner -> Change. You need administrative privileges to do this.

18. What are "inherited permissions"?

A: Permissions that are propagated to a child object (file or subfolder) from its parent object (folder).

19. Describe a scenario where you would break inheritance and apply unique permissions to a subfolder.

A: A shared "Projects" folder is accessible by all departments. Inside it, a "HR-Salaries" subfolder should only be accessible by the HR group. You would break inheritance on "HR-Salaries" and grant permissions only to the HR group.

20. What is the purpose of the "Public" folder in Windows, and how is its sharing configured?

A: It's a default folder for easy file sharing. Sharing is turned on or off for all users via Network and Sharing Center -> "Public folder sharing".

Post-Lab Activity 13: Apply the concept of networking to understand Virtual LAN Networks and configure VLAN.

Basic (1-10)

1. What does VLAN stand for?

A: Virtual Local Area Network.

2. What is the primary purpose of a VLAN?

A: To logically segment a physical network into multiple, isolated broadcast domains.

3. What problem do VLANs solve?

A: They reduce network congestion by containing broadcast traffic within a smaller group of devices.

4. Can devices on different VLANs communicate with each other by default?

A: No. Communication between VLANs requires a router (Layer 3 device).

5. What is an "access port" on a switch?

A: A port that carries traffic for only one specific VLAN.

6. What is a "trunk port" on a switch?

A: A port that carries traffic for multiple VLANs, typically between switches or to a router.

7. What protocol is commonly used to manage VLAN trunks between switches?

A: IEEE 802.1Q.

8. What is the "native VLAN"?

A: The VLAN on a trunk port that carries untagged traffic.

9. Give an example of why you would put devices in separate VLANs.

A: To separate guest WiFi traffic from the corporate network for security.

10. What is the default VLAN on most switches?

A: VLAN 1.

Intermediate (11-15)

11. What is the main difference between a physical LAN and a VLAN?

A: A physical LAN is defined by physical switch connections. A VLAN is a logical segmentation created by software configuration, independent of physical location.

12. How does a switch know which VLAN a frame belongs to on a trunk link?

A: Through an 802.1Q tag added to the Ethernet frame header, which contains the VLAN ID (VID).

13. Why is it important to change the native VLAN from the default (VLAN 1) for security?

A: VLAN 1 is well-known and often targeted. Changing it to an unused VLAN can help mitigate VLAN hopping attacks.

14. What is "VLAN hopping" and how is it prevented?

A: An attack where a user's traffic is sent to a VLAN they are not part of. It is prevented by explicitly setting the native VLAN on trunks to an unused VLAN and not using it for user access ports.

15. What is the command to create a VLAN (e.g., VLAN 10) on a Cisco switch?

A: vlan 10

Advanced (16-20)

16. Describe the step-by-step process of configuring a switch port as an access port for VLAN 20.

A: (Cisco Example)

1. configure terminal

2. interface fastethernet0/1

3. switchport mode access

4. switchport access vlan 20

5. end

17. How do you configure a switch port as a trunk port?

A: (Cisco Example)

a. interface gigabitethernet0/1
b. switchport mode trunk
c. switchport trunk native vlan 99 (Best practice)
d. switchport trunk allowed vlan 10,20,30 (Optional, for security)

18. Explain the concept of "router-on-a-stick."

A: A network configuration where a single physical router interface is subdivided into multiple logical sub-interfaces, each acting as the default gateway for a different VLAN. The physical port is connected to a switch trunk.

19. What is the purpose of the show vlan brief command?

A: To display a summary of all configured VLANs and the access ports assigned to them.

20. How can VLANs be extended across multiple physical switches?

A: By connecting the switches via trunk ports. The VLANs are created on both switches, and the trunk link carries the traffic for all those VLANs between them.